Project Title: ***Malaria Detection using Deep Learning*** **Project Design Phase-I - Solution Fit Template** **Team ID:** PNT2022TMIDxxxxxx

Define CS, fit into CC

1. CUSTOMER SEGMENT(S) CS

Who is your customer?
i.e. working parents of 0-5 y.o. kids

- Hospitals and diagnostic laboratories.
- Pathologists and lab technicians analysing blood smear images.
- Healthcare organizations in malaria-prone regions.
- Government health departments and medical research institutes.
- Rural healthcare centres with limited diagnostic

6. CUSTOMER CONSTRAINTS CC

- Limited access to advanced diagnostic technologies in rural areas.
- Lack of trained medical professionals in some regions.
- Budget constraints in small healthcare facilities.
- Limited computing infrastructure in rural clinics.
- Dependence on manual microscopy for diagnosis.

5. AVAILABLE SOLUTIONS AS

Currently malaria detection is mainly performed through:

- Manual microscopic examination of blood smear slides.
- Rapid Diagnostic Tests (RDTs).
- Polymerase Chain Reaction (PCR) based laboratory tests.

Limitations of these solutions:

- Time-consuming process.
- Requires skilled laboratory professionals.

Explore AS, differentiate

2. JOBS-TO-BE-DONE / PROBLEMS

Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides.

- Detect malaria parasites in microscopic blood smear images.
- Reduce time required for manual microscopic diagnosis.
- Minimize human errors during parasite identification.
- Enable faster diagnosis for early treatment of malaria patients.
- Provide affordable diagnostic support for resource-limited healthcare facilities.

9. PROBLEM ROOT CAUSE RC

- Manual malaria diagnosis requires highly skilled technicians.
- Large number of samples increases workload in laboratories.
- Lack of automated tools for parasite detection in blood smear images.
- Delays in diagnosis can delay treatment for patients.
- Limited access to advanced diagnostic systems in remote areas.

7. BEHAVIOUR BE

- Doctors examine blood smear images using microscopes.
- Lab technicians manually count and analyze infected cells.
- Healthcare workers send samples to diagnostic laboratories.
- Hospitals maintain records of malaria tests for patient monitoring.
- Medical professionals rely on experience and laboratory reports for diagnosis.

Identify strong TR & EM

3. TRIGGERS TR

What triggers customers to act? i.e. seeing their neighbor installing solar panels, reading about a more efficient solution in the news.

Increase in cases of Malaria in certain regions.

Need for faster and more reliable diagnostic tools in healthcare.

Adoption of artificial intelligence in medical image analysis.

Demand for automated systems that assist doctors and lab technicians.

Public health initiatives promoting early malaria detection.

10. YOUR SOLUTION SL

Our project proposes an AI-based Malaria Detection System that automatically classifies microscopic blood smear images as Parasitized or Uninfected using deep learning techniques.

The system uses multiple deep learning models including:

- Custom Convolutional Neural Network (CNN)
- MobileNetV2 Transfer Learning Model
- EfficientNetB0 Transfer Learning Model

The trained model is integrated into a Flask-based web application where users can upload blood smear images. The system processes the image using the trained model and instantly predicts whether malaria parasites are present.

Benefits of the solution:

- Faster malaria diagnosis.

8. CHANNELS of BEHAVIOUR CH

8.1 ONLINE

- Web-based healthcare diagnostic tools.
- Medical research databases and digital healthcare platforms.
- Telemedicine services and cloud-based diagnostic systems.
- Online medical imaging analysis tools.

8.2 OFFLINE

- Hospitals and diagnostic laboratories.
- Microscopic examination of blood samples.
- Rural healthcare camps and clinics.
- Government health programs for malaria screening.

Identify strong TR & EM

4. EMOTIONS: BEFORE / AFTER EM

How do customers feel when they face a problem or a job and afterwards? i.e. lost, insecure > confident, in control - use it in your communication strategy & design.4

Before using the solution

- Doctors and technicians feel stressed due to heavy workload.
- Patients feel anxious waiting for test results.
- Risk of misdiagnosis due to manual examination.

After using the solution

- Doctors feel confident with AI-assisted diagnosis.
- Faster results improve patient satisfaction.
- Healthcare workers feel relieved with reduced workload.
- Reduced workload for lab technicians.
- Improved diagnostic accuracy using AI.
- Affordable and accessible screening tool for healthcare facilities.